



DATASET PROFILING & SUMMARY STATS REPORT

DATASET USED: ADULT PSYCHIATRIC MORBIDITY SURVEY (APMS) 2014

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DATE: MAY 3, 2026

Introduction:

The dataset (APMS – Annual Status of Education Report type data) contains structured information collected from surveys. It includes multiple fields such as demographic details, education-related indicators, and categorical/numerical variables.

- the data is stored in csv (comma-separated values) format
- each row represents a record/observation
- each column represents a feature/attribute
- the dataset may include:
 - o numeric data (e.g., age, scores)
 - o categorical data (e.g., gender, category)
 - o missing values (empty or null fields)

TECHNOLOGY USED

- **Python** – Programming language for analysis
- **Jupyter Notebook** – Interactive environment for coding and visualization
- **Pandas** – Data manipulation and analysis library

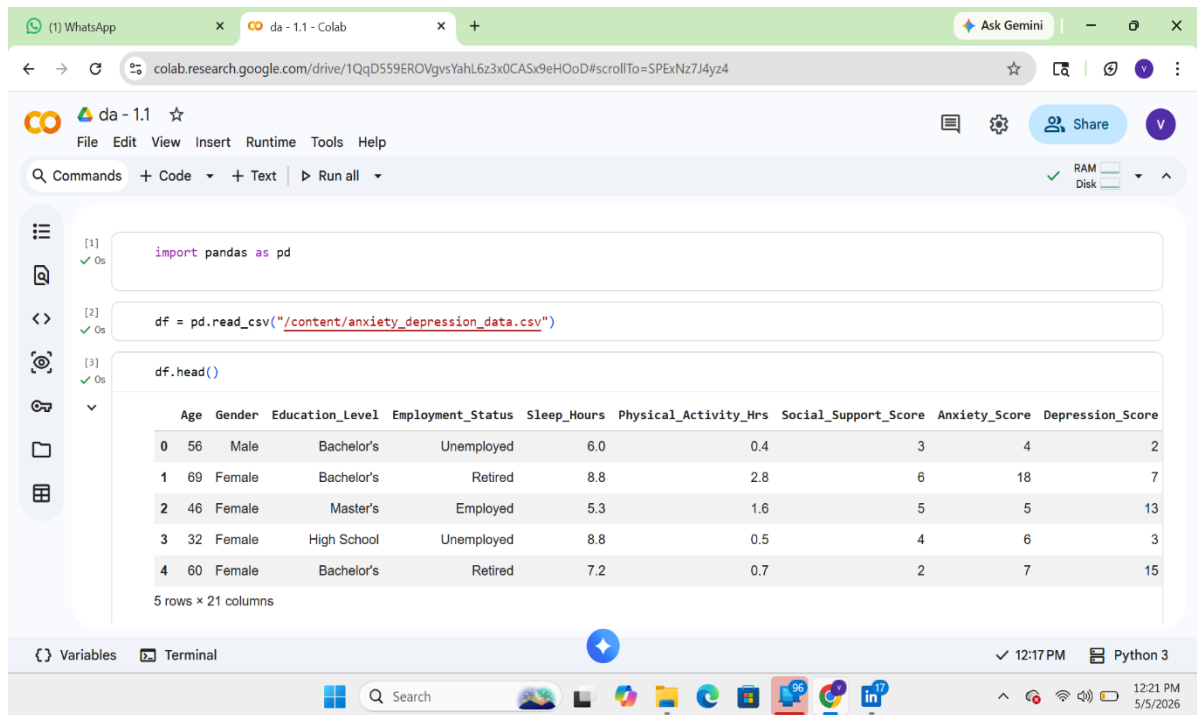
DATASET PROFILING:

Dataset profiling is the process of examining the dataset to understand its quality and characteristics.

Row Counts:

- Total number of records in the dataset

- Helps understand dataset size and completeness



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da - 1.1

File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

[1] ✓ Os

```
import pandas as pd
```

[2] ✓ Os

```
df = pd.read_csv("/content/anxiety_depression_data.csv")
```

[3] ✓ Os

```
df.head()
```

	Age	Gender	Education_Level	Employment_Status	Sleep_Hours	Physical_Activity_Hrs	Social_Support_Score	Anxiety_Score	Depression_Score
0	56	Male	Bachelor's	Unemployed	6.0	0.4	3	4	2
1	69	Female	Bachelor's	Retired	8.8	2.8	6	18	7
2	46	Female	Master's	Employed	5.3	1.6	5	5	13
3	32	Female	High School	Unemployed	8.8	0.5	4	6	3
4	60	Female	Bachelor's	Retired	7.2	0.7	2	7	15

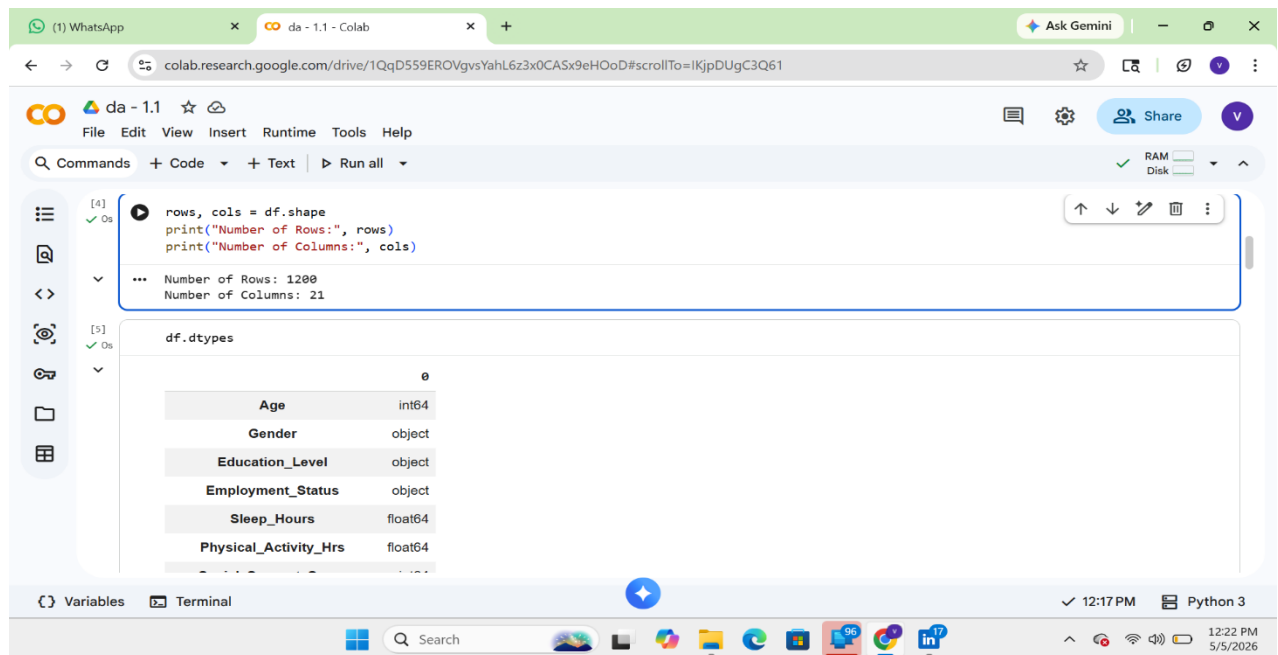
5 rows × 10 columns

Variables Terminal

12:17 PM Python 3

Search

12:21 PM 5/5/2026



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File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

[4] ✓ Os

```
rows, cols = df.shape
print("Number of Rows:", rows)
print("Number of Columns:", cols)
```

... Number of Rows: 1200
Number of Columns: 21

[5] ✓ Os

```
df.dtypes
```

Age	int64
Gender	object
Education_Level	object
Employment_Status	object
Sleep_Hours	float64
Physical_Activity_Hrs	float64

Variables Terminal

12:17 PM Python 3

Search

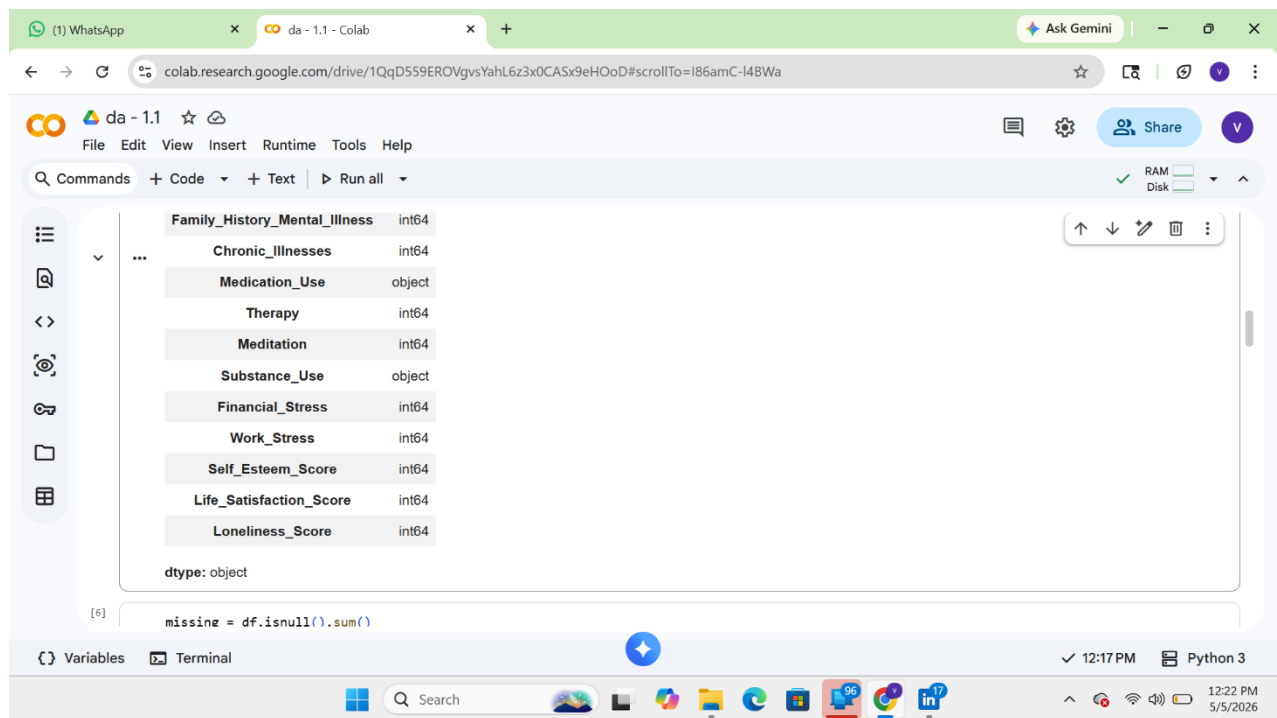
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FIELD DATATYPES:

- Identifies whether columns are:
 1. Numerical (int, float)
 2. Categorical (object/string)

The screenshot shows a Google Colab notebook interface. The top bar includes a search bar with "Ask Gemini" and a "Share" button. The notebook's menu bar shows "File", "Edit", "View", "Insert", "Runtime", "Tools", and "Help". Below the menu, the "Commands" bar has options for "Code", "Text", and "Run all". The main workspace displays the output of the command `df.dtypes`, which is a table showing the data types for each column in a DataFrame. The table has two columns: the column name and its corresponding data type. The data types are: Age (int64), Gender (object), Education_Level (object), Employment_Status (object), Sleep_Hours (float64), Physical_Activity_Hrs (float64), Social_Support_Score (int64), Anxiety_Score (int64), Depression_Score (int64), Stress_Level (int64), and Family_History_Mental_Illness (int64). The bottom of the interface shows a "Variables" tab, a "Terminal" tab, and a system tray with the time "12:17 PM" and "Python 3".

Age	int64
Gender	object
Education_Level	object
Employment_Status	object
Sleep_Hours	float64
Physical_Activity_Hrs	float64
Social_Support_Score	int64
Anxiety_Score	int64
Depression_Score	int64
Stress_Level	int64
Family_History_Mental_Illness	int64



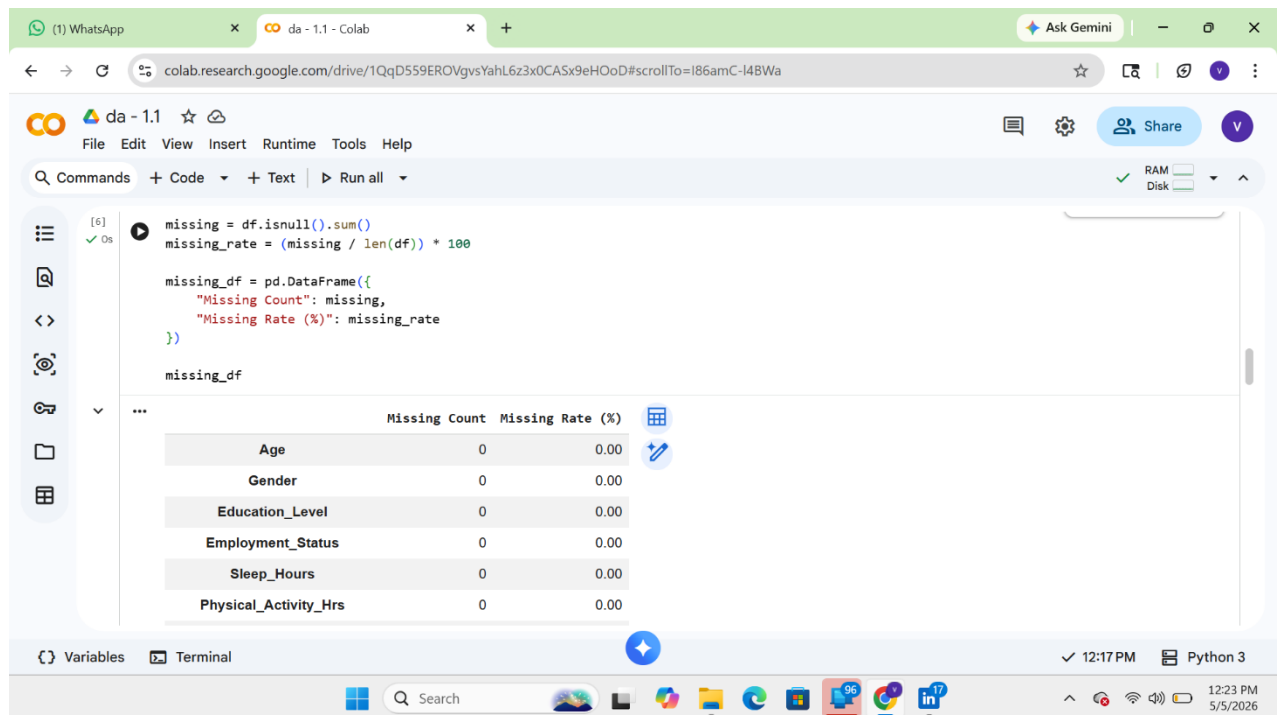
The screenshot shows a Google Colab notebook interface. The top bar includes a WhatsApp icon, a tab for 'da - 1.1 - Colab', and a 'Ask Gemini' button. The address bar shows the Colab URL. The notebook's menu bar includes 'File', 'Edit', 'View', 'Insert', 'Runtime', 'Tools', and 'Help'. Below the menu is a toolbar with 'Commands', '+ Code', '+ Text', and 'Run all'. The main area displays a pandas DataFrame with the following columns and dtypes:

Column Name	Dtype
Family_History_Mental_Illness	int64
Chronic_Illnesses	int64
Medication_Use	object
Therapy	int64
Meditation	int64
Substance_Use	object
Financial_Stress	int64
Work_Stress	int64
Self_Esteem_Score	int64
Life_Satisfaction_Score	int64
Loneliness_Score	int64

The DataFrame's dtype is 'object'. Below the DataFrame, the code cell shows the command `missing = df.isnull().sum()`. The bottom status bar indicates 'Variables', 'Terminal', '12:17 PM', and 'Python 3'.

MISSING VALUES:

- Detects null or missing values in each column
- Helps in data cleaning decisions (remove/fill values)



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```
[6] ✓ 0s
missing = df.isnull().sum()
missing_rate = (missing / len(df)) * 100

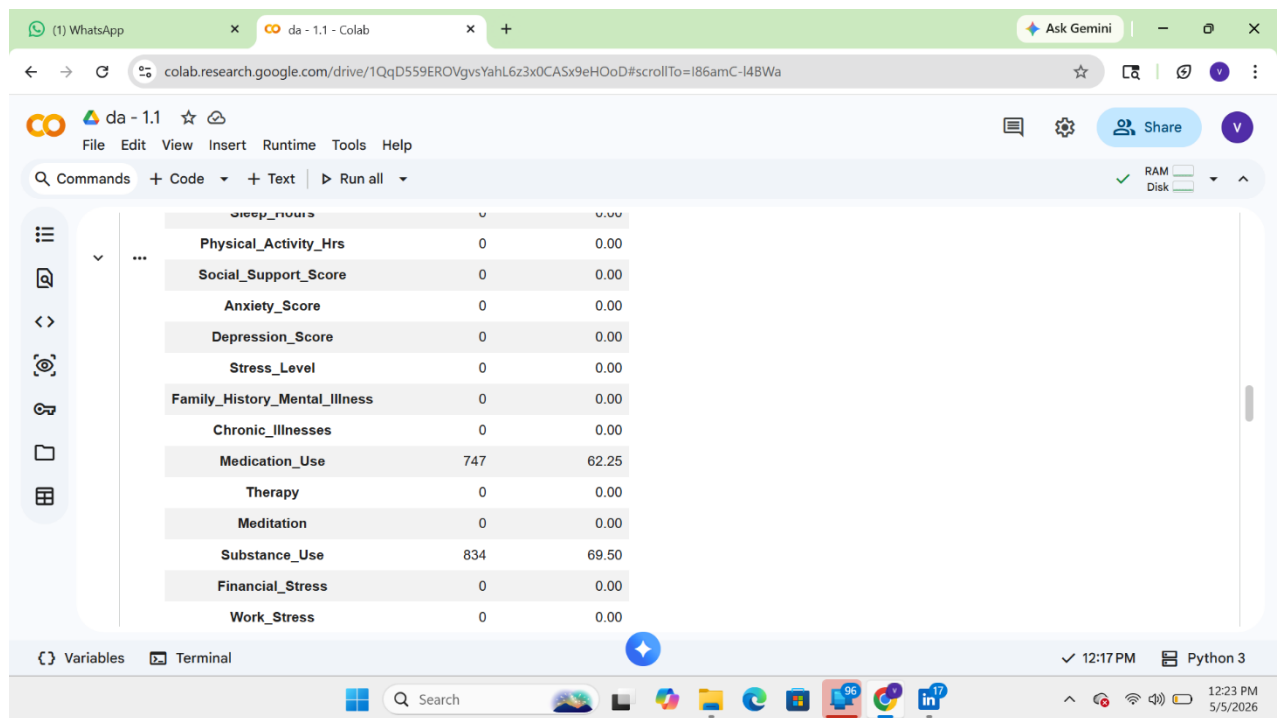
missing_df = pd.DataFrame({
    "Missing Count": missing,
    "Missing Rate (%)": missing_rate
})

missing_df
```

The output of the code cell is a DataFrame with the following columns and values:

	Missing Count	Missing Rate (%)
Age	0	0.00
Gender	0	0.00
Education_Level	0	0.00
Employment_Status	0	0.00
Sleep_Hours	0	0.00
Physical_Activity_Hrs	0	0.00

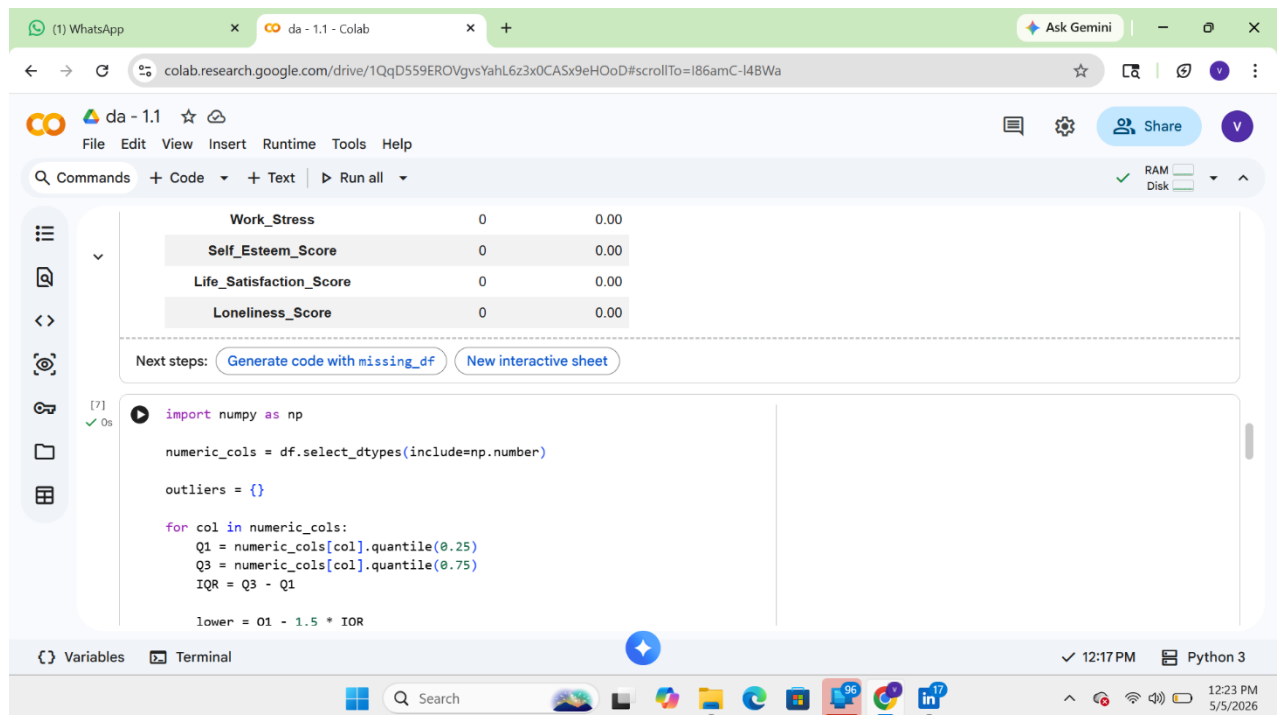
The bottom status bar indicates 'Variables', 'Terminal', '12:17 PM', and 'Python 3'.



The screenshot shows a Google Colab notebook interface. The top bar includes a WhatsApp icon, a tab for 'da - 1.1 - Colab', and a 'Ask Gemini' button. The address bar shows the URL: colab.research.google.com/drive/1QqD559EROVgvsYahL6z3x0CAsx9eHOoD#scrollTo=I86amC-I48Wa. The notebook interface shows a menu bar (File, Edit, View, Insert, Runtime, Tools, Help) and a toolbar with 'Commands', '+ Code', '+ Text', and 'Run all'. On the left, there is a sidebar with icons for file management. The main area displays a table of variables and their values:

Variable	Value	Value
Physical_Activity_Hrs	0	0.00
Social_Support_Score	0	0.00
Anxiety_Score	0	0.00
Depression_Score	0	0.00
Stress_Level	0	0.00
Family_History_Mental_Illness	0	0.00
Chronic_Illnesses	0	0.00
Medication_Use	747	62.25
Therapy	0	0.00
Meditation	0	0.00
Substance_Use	834	69.50
Financial_Stress	0	0.00
Work_Stress	0	0.00

The bottom bar shows 'Variables', 'Terminal', and the time '12:17 PM' with 'Python 3' selected.



The screenshot shows the same Google Colab notebook interface, but now with code for outlier detection. The top bar and address bar are identical. The notebook interface shows the same menu bar and toolbar. The main area displays the following code:

```
[7]
import numpy as np

numeric_cols = df.select_dtypes(include=np.number)

outliers = {}

for col in numeric_cols:
    Q1 = numeric_cols[col].quantile(0.25)
    Q3 = numeric_cols[col].quantile(0.75)
    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
```

Below the code, there are two buttons: 'Generate code with missing_df' and 'New interactive sheet'. The bottom bar shows 'Variables', 'Terminal', and the time '12:17 PM' with 'Python 3' selected.

OUTLIER DETECTION:

- Identifies extreme values using methods like **IQR (Interquartile Range)**
- Important for improving model accuracy and avoiding bias

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```
import numpy as np

numeric_cols = df.select_dtypes(include=np.number)

outliers = {}

for col in numeric_cols:
    Q1 = numeric_cols[col].quantile(0.25)
    Q3 = numeric_cols[col].quantile(0.75)
    IQR = Q3 - Q1

    lower = Q1 - 1.5 * IQR
    upper = Q3 + 1.5 * IQR

    outliers[col] = ((numeric_cols[col] < lower) | (numeric_cols[col] > upper)).sum()

outliers_df = pd.DataFrame.from_dict(outliers, orient='index', columns=['Outlier Count'])
outliers_df
```

Outlier Count

Variables Terminal

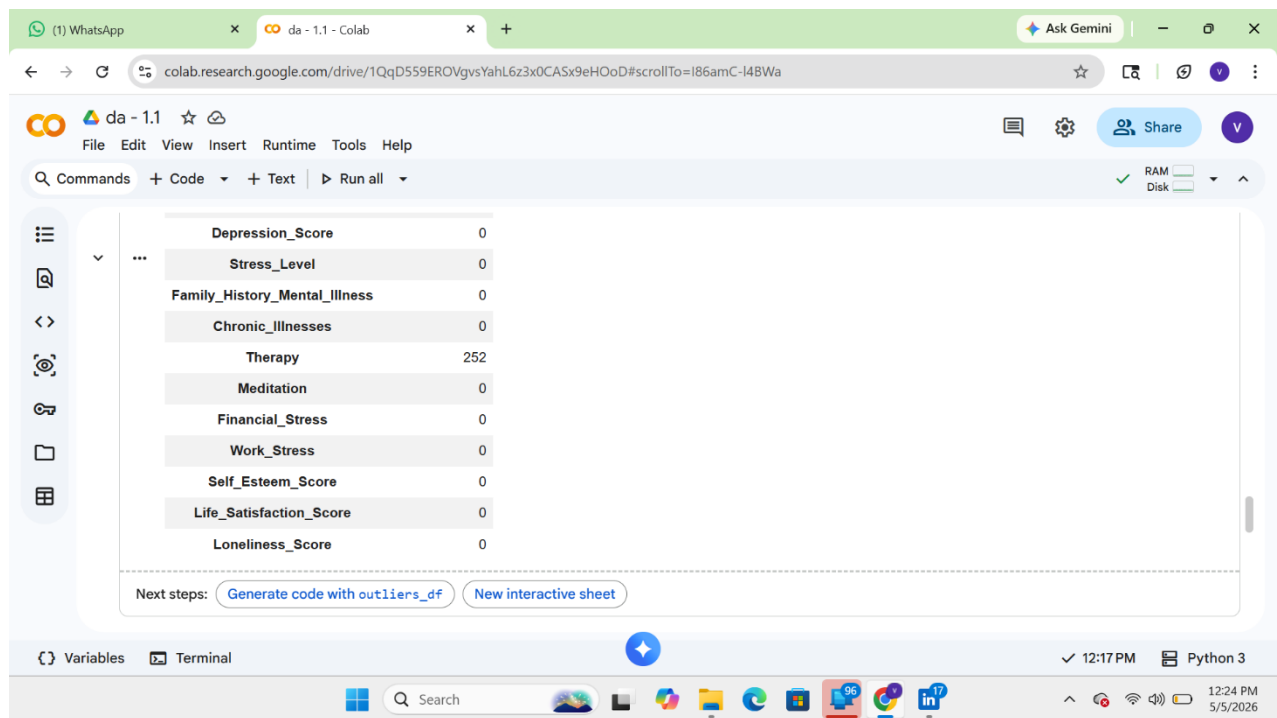
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	Outlier Count
Age	0
Sleep_Hours	6
Physical_Activity_Hrs	75
Social_Support_Score	0
Anxiety_Score	0
Depression_Score	0
Stress_Level	0
Family_History_Mental_Illness	0
Chronic_Illnesses	0
Therapy	252
Meditation	0

Variables Terminal

12:17 PM Python 3



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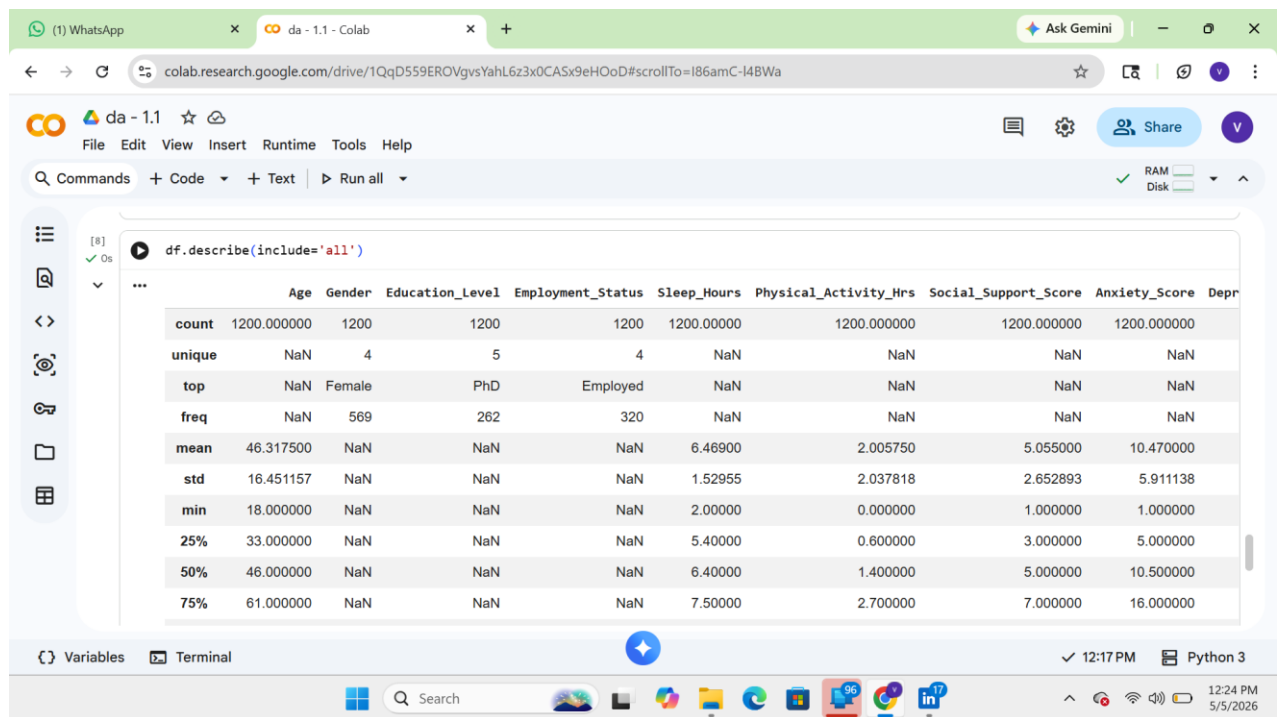
RAM Disk

Depression_Score	0
Stress_Level	0
Family_History_Mental_Illness	0
Chronic_Illnesses	0
Therapy	252
Meditation	0
Financial_Stress	0
Work_Stress	0
Self_Esteem_Score	0
Life_Satisfaction_Score	0
Loneliness_Score	0

Next steps: [Generate code with outliers_df](#) [New interactive sheet](#)

Variables Terminal 12:17 PM Python 3

Output descriptive stats table:



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File Edit View Insert Runtime Tools Help

Commands + Code + Text Run all

RAM Disk

```
[8] df.describe(include='all')
```

	Age	Gender	Education_Level	Employment_Status	Sleep_Hours	Physical_Activity_Hrs	Social_Support_Score	Anxiety_Score	Depression_Score
count	1200.000000	1200	1200	1200	1200.000000	1200.000000	1200.000000	1200.000000	1200.000000
unique	NaN	4	5	4	NaN	NaN	NaN	NaN	NaN
top	NaN	Female	PhD	Employed	NaN	NaN	NaN	NaN	NaN
freq	NaN	569	262	320	NaN	NaN	NaN	NaN	NaN
mean	46.317500	NaN	NaN	NaN	6.46900	2.005750	5.055000	10.470000	NaN
std	16.451157	NaN	NaN	NaN	1.52955	2.037818	2.652893	5.911138	NaN
min	18.000000	NaN	NaN	NaN	2.000000	0.000000	1.000000	1.000000	NaN
25%	33.000000	NaN	NaN	NaN	5.400000	0.600000	3.000000	5.000000	NaN
50%	46.000000	NaN	NaN	NaN	6.400000	1.400000	5.000000	10.500000	NaN
75%	61.000000	NaN	NaN	NaN	7.500000	2.700000	7.000000	16.000000	NaN

Variables Terminal 12:17 PM Python 3

	mean	std	min	25%	50%	75%	max
std	16.451157	NaN	NaN	NaN	1.52955	2.037818	2.652893
min	18.000000	NaN	NaN	NaN	2.000000	0.000000	1.000000
25%	33.000000	NaN	NaN	NaN	5.400000	0.600000	3.000000
50%	46.000000	NaN	NaN	NaN	6.400000	1.400000	5.000000
75%	61.000000	NaN	NaN	NaN	7.500000	2.700000	7.000000
max	74.000000	NaN	NaN	NaN	12.400000	15.100000	9.000000

11 rows x 21 columns

CONCLUSION :

- the dataset contains structured survey data stored in csv format • python with jupyter notebook is used for analysis
- pandas helps in data cleaning and manipulation
- Understanding the dataset clearly
- Detecting data quality issues
- Data issues are identified quickly
- Summary statistics are generated efficiently
- Outliers and missing values are detected accurately